Samson Lai

AI Engineer | RAG & Agentic Systems | Backend Engineering

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Professional Summary

AI-focused Computer Science graduate (First Class Honours) with backend engineering experience and hands-on development of deployed RAG, agentic, and collaborative AI systems. Builds Python/FastAPI services, hybrid retrieval pipelines, LLM agents, and full-stack AI products with PostgreSQL/pgvector, LangGraph, Supabase, Trigger.dev, Docker, Cloudflare, and other cloud platforms.

Professional Experience

Business Analyst, Hilti Aug 2024 – Nov 2024

- Supported Salesforce CRM operations for global sales teams by configuring region-specific functionality and Salesforce Flow automations based on business requirements and local policies.
- Developed conditional Salesforce Flow workflows, including follow-up reminders triggered by customer inactivity, to support lead and customer relationship management.
- Supported the requirements-to-release lifecycle by analyzing ServiceNow requests and user feedback, translating them into Jira tickets, resolving assigned issues, and testing completed changes before release.

Backend Developer, Foodport Dec 2023 – June 2024

- Developed RESTful CRUD APIs for a social food-recommendation platform using Node.js, Express.js, and MongoDB.
- Implemented OAuth 2.0 and Firebase Authentication for protected API access, and verified endpoint behavior with Postman and scripted tests covering success and error cases.
- Containerized backend services with Docker, managed source control with Git, and deployed the application to AWS.

Education

Asia Pacific University of Technology & Innovation (APU), Bachelor of Computer Science (Honours) in Artificial Intelligence - First Class Honours 2022 – 2025

- CGPA: 3.66/4.00

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, Java

AI / LLM: RAG, LangGraph, LangChain, MCP, OpenAI API, Gemini, LiteLLM, PydanticAI, tool calling, hybrid retrieval, RRF

Backend / Data: FastAPI, Node.js, Express.js, PostgreSQL, pgvector, Supabase, Qdrant, ChromaDB, MongoDB, SQLite, SQLAlchemy, Alembic, Prisma

Frontend / Realtime: React, Next.js, React Flow, Liveblocks

Cloud / DevOps / Quality: Cloudflare, Docker, AWS, Railway, Vercel, Trigger.dev, Git/GitHub, GitHub Actions, pytest, Postman, LangSmith

Projects

FilingLens - Grounded RAG Research Platform for SEC Filings 2026

- Implemented a deployed RAG application over 25 SEC 10-K filings across five public companies, combining pgvector semantic search and PostgreSQL full-text search with Reciprocal Rank Fusion to retrieve relevant financial-document evidence.
- Enforced grounded answers with source-level citations and a fail-closed validator that rejects citations not backed by retrieved passages.
- Shipped an authenticated FastAPI + React/TypeScript system with Supabase PostgreSQL/Auth, SQLAlchemy/Alembic, OpenAI, pytest unit/integration tests, and separate Railway deployments.

StackCanvas - Collaborative AI System Design Platform 2026

GitHub

- Built a real-time collaborative architecture workspace with Next.js, TypeScript, React Flow, and Liveblocks, supporting synchronized editing, presence, and live cursors.
- Automated natural-language architecture generation and technical-spec creation with Gemini and Trigger.dev durable background workflows, translating requirements into structured canvas nodes, edges, and persisted specifications.
- Shipped authenticated project ownership and collaborator access with Clerk, Prisma/PostgreSQL, Vercel Blob, and Vercel deployment.

Autonomous Support Engine - Agentic Customer Support System 2026

GitHub

- Built a stateful LangGraph agent with separate triage, resolution, and QA stages, category-aware tool dispatch, and a three-iteration circuit breaker to bound autonomous loops.
- Added a dual-model QA gate using Gemini for resolution and local Qwen for review, with ChromaDB RAG and policy checks that reject hallucinated, over-limit, or low-quality responses.
- Instrumented the agent lifecycle with SQLite checkpointing, LangSmith tracing, MCP tool integration, and a golden-dataset evaluation harness across billing, technical, and general support scenarios.

Honors and Awards

Hilti ITC 2024 - Finalist: Developed an AI-powered HR system focused on enterprise talent acquisition.

3rd Prize - Medical Grand Challenge: Built a healthcare technology solution addressing a real-world medical challenge.